

Fullstack Development

Preflight project - frontend

[Github Repo](#)

Setup (Repos)

- DB
- Backend

Frontend framework

Tool that enables you to develop the frontend of your applications

Why?

- You don't have to everything code from scratch.
- You can collaborate with others more easily.

Frontend JS framework

- [State of JS 2025](#)
- [Stack Overflow Survey 2025](#)

Frontend tooling

- Code bundler ("*build tools*")
 - Optimizes your code for production
- Development server
 - Hot module replacement (HMR)
- Optimizing production builds
- Support
 - TypeScript, JSX, CSS preprocessors, assets, plugins, etc.

Frontend tooling

- Vite
- Webpack
- Parcel
- Rollup

Setup

- React
 - `pnpm create vite@latest` (*Follow the instruction*)
- Additional libraries
 - `pnpm i dayjs axios @picocss/pico`

Files

- Remove all `css` files
- `./vite.config.ts` [\(Link\)](#)
- `./src/main.ts` [\(Link\)](#)
- `./src/types.ts` [\(Link\)](#)
 - I use [Quicktype](#) to generate this.

Minimal example

- `./src/App.tsx` [\(Link\)](#)
- Start
 - `pnpm run dev`

Full example

- `./src/App.tsx` [\(Link\)](#)
- Build
 - `pnpm run build`
- Preview
 - `pnpm run preview`

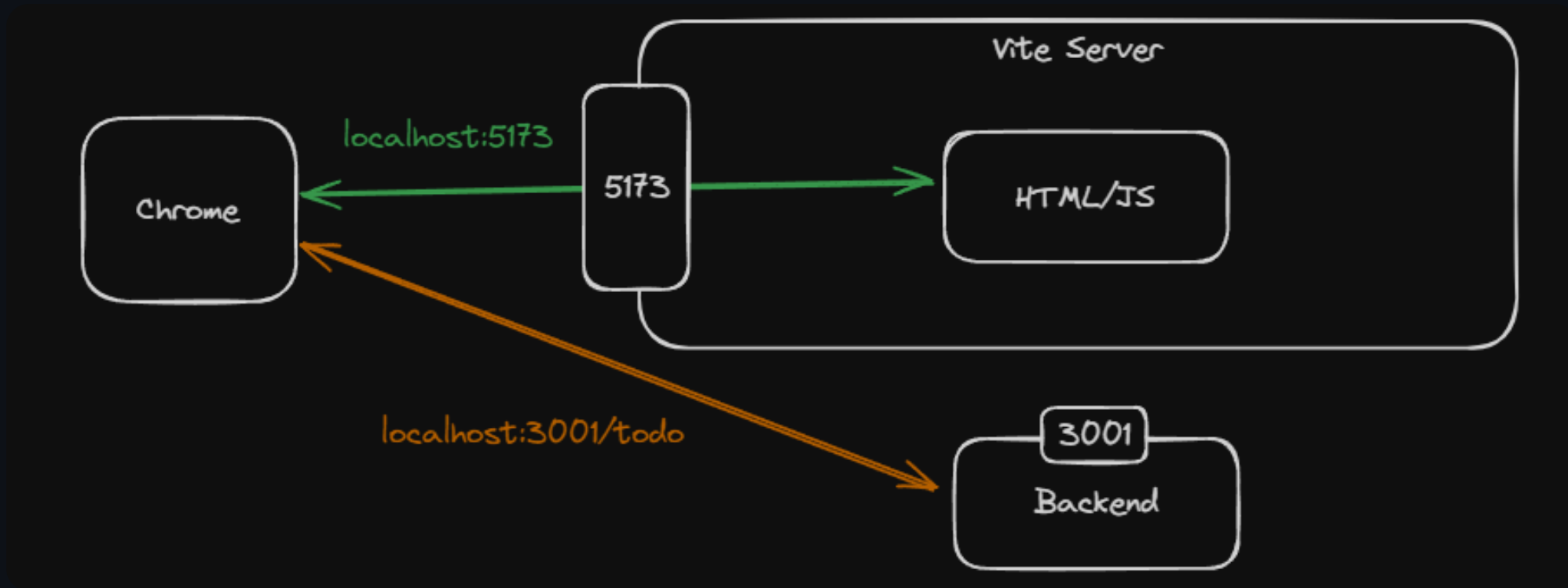
CORS

- Cross-Origin Resource Sharing
- A browser security feature
 - Restricts **cross-origin** HTTP requests initiated from scripts running in the browser.
 - `localhost:5173` (frontend) → `localhost:3001` (backend)
- The backend must explicitly allow the frontend to access its resources.
- Solution: **reverse proxy**.

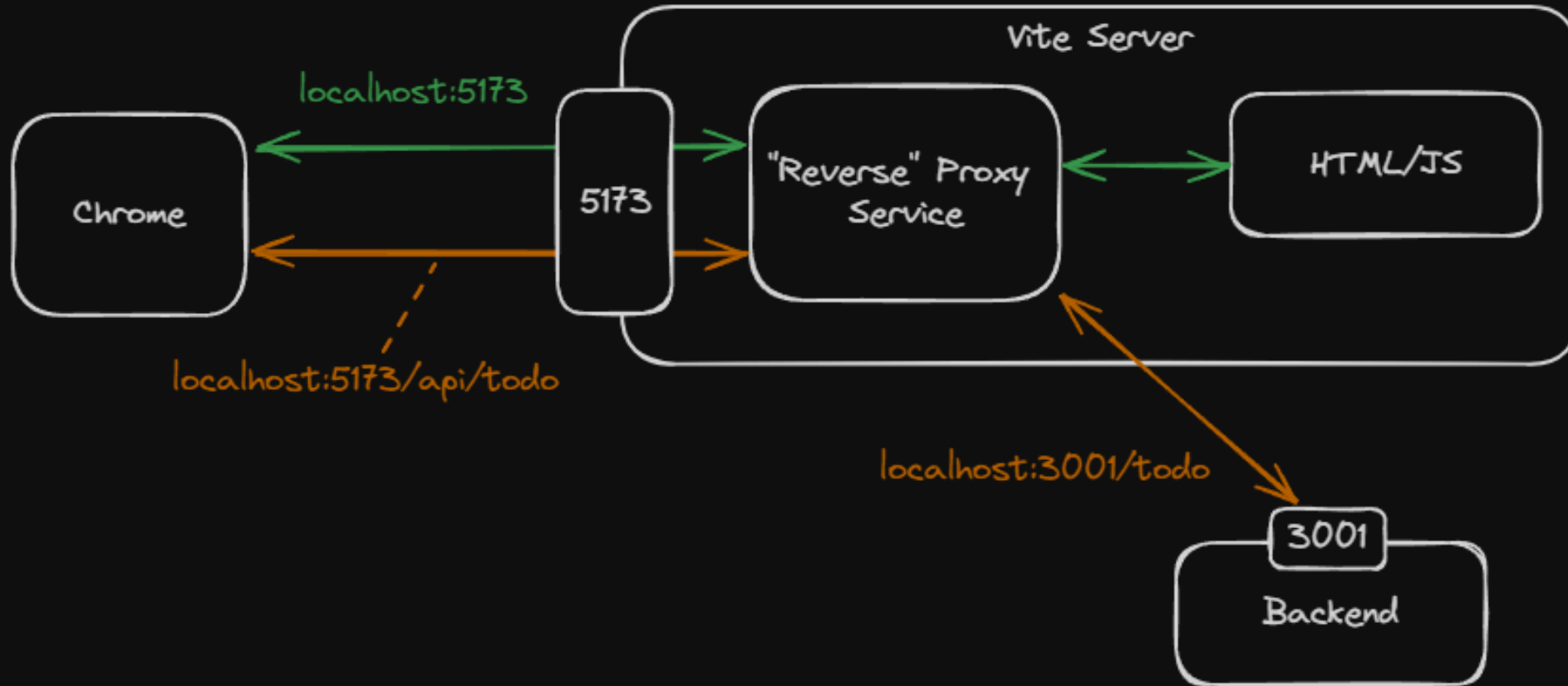
Why reverse proxy?

- No issues with CORS and cookies.
- No need to expose your backend to the internet.

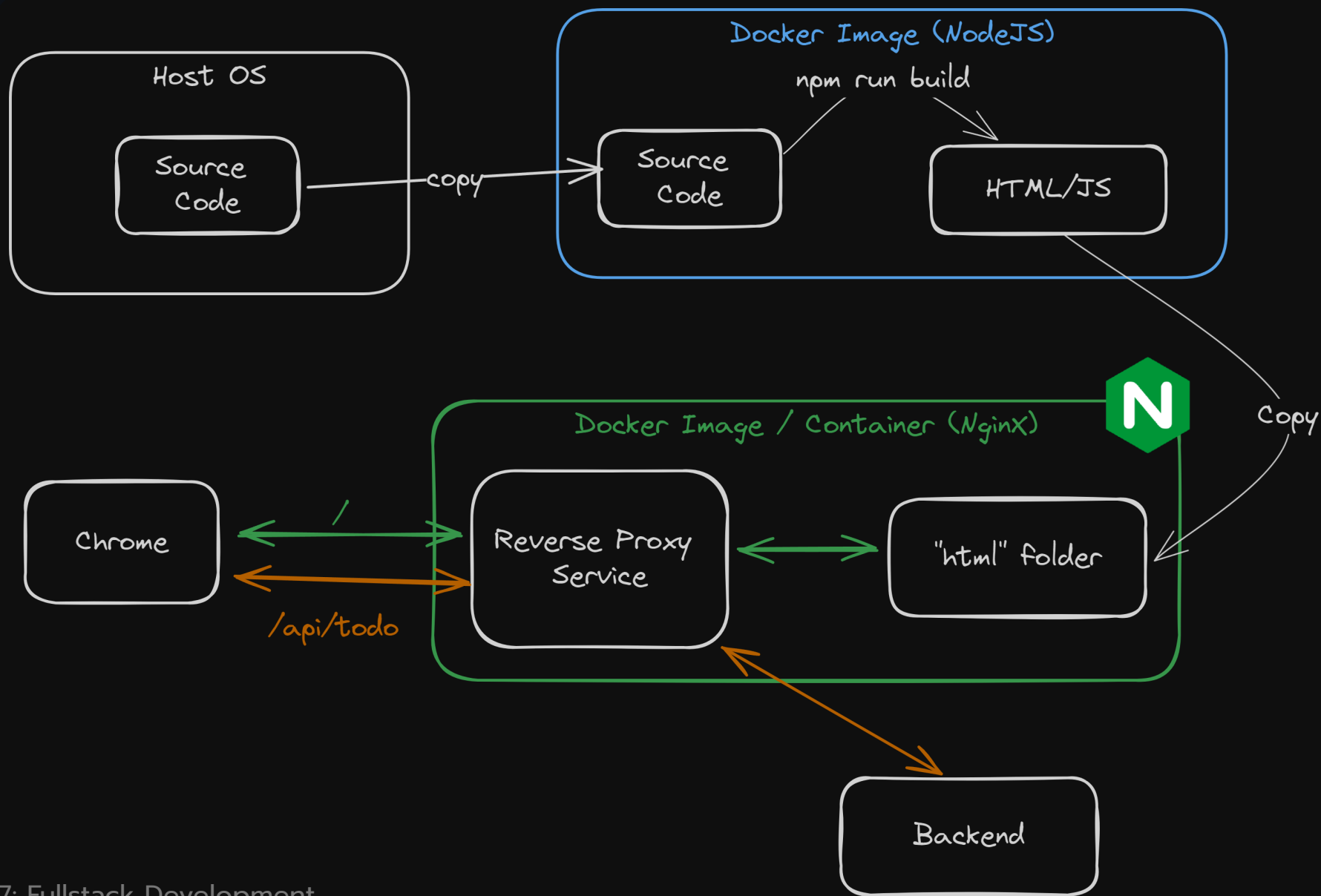
Without reverse proxy



With reverse proxy

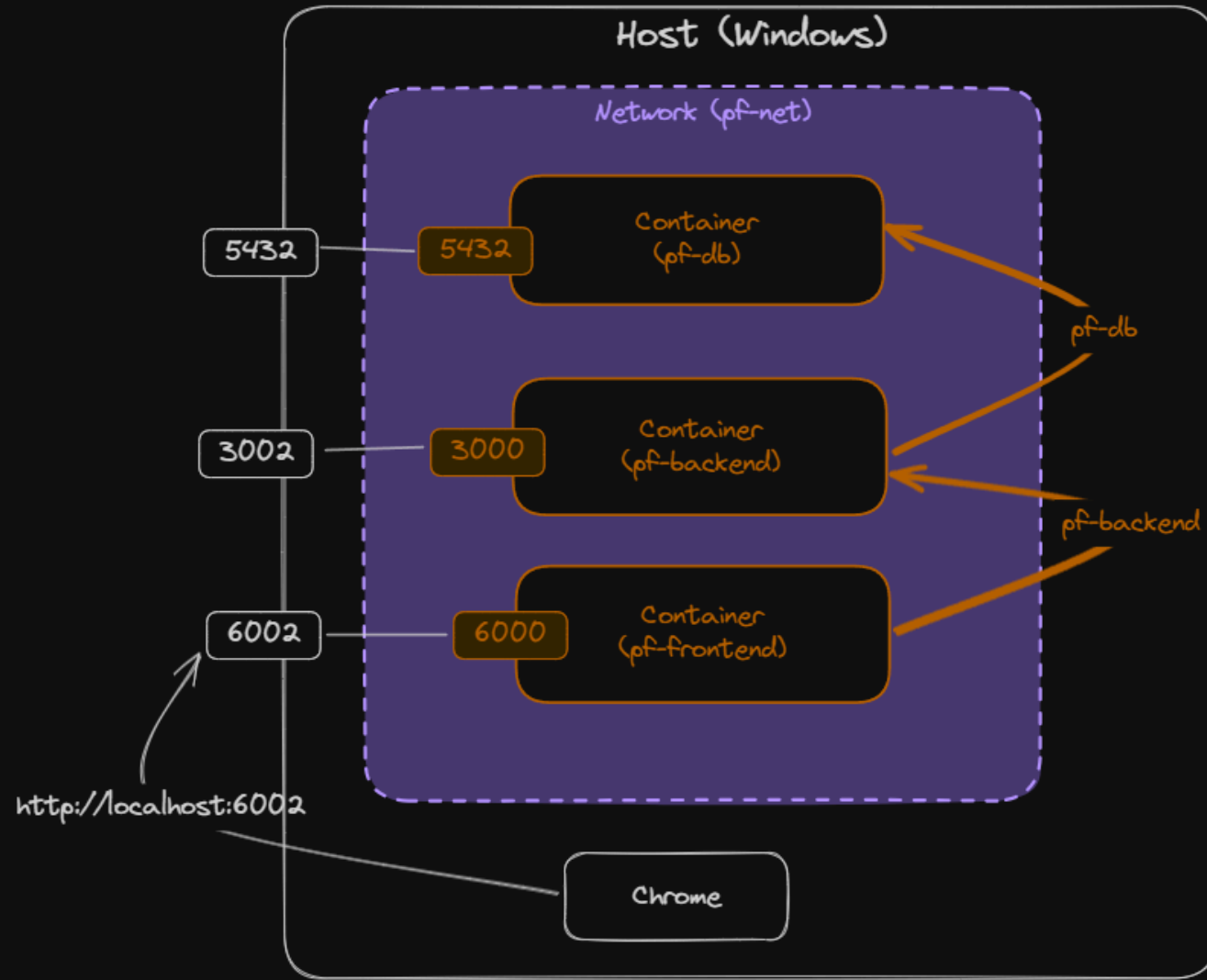


Containerization



Steps

-  Make `./.env.test` from `./.env.test.example` ([Link](#))
-  `./Dockerfile` ([Link](#))
-  `./.dockerignore` ([Link](#))
-  `./nginx.conf.template` ([Link](#)) ([Why](#))
-  `./docker-compose.yml` ([Link](#))
-  `docker compose --env-file ./.env.test up -d --force-recreate --build`



Push to Dockerhub

- Create an account at <https://hub.docker.com>.
- Create repository called `preflight-frontend`.
- Tag image
 -  `docker tag preflight-frontend [DOCKERHUB_ACCOUNT]/preflight-frontend:latest`
- Login (if you have not done so)
 -  `docker login`
- Push image
 -  `docker push [DOCKERHUB_ACCOUNT]/preflight-frontend:latest`